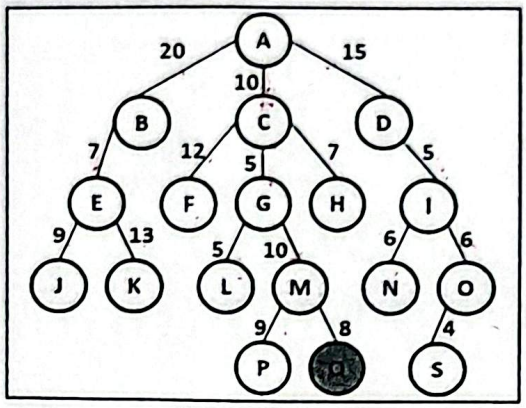
Solution

Artificial Intelligence – BSCS – Spring 2026
Quiz 2

Name: _____ **Reg.No.** _____

Question-4: Informed Searches **[15+5]**

The **start node is A**, and the **goal node is Q**. The number on the edges indicate the associated **path cost i.e. g(n)**. The **heuristic value h(n)** of each node is given in the table.



State	h(n)	State	h(n)
A	40	K	90
B	60	L	70
C	70	M	100
D	65	N	130
E	70	O	120
F	90	P	110
G	110	Q	0
H	80	R	120
I	90	S	150

a. Perform the A* algorithm on the above graph.

b. Is **heuristic admissible**? Give reasons to support your choice.

1. (A, 40) — A
2. ~~(B, 80)~~ (C, 87) (D, 80) — B
3. ~~(C, 80)~~ (D, 80) (E, 97) — C
4. ~~(D, 80)~~ (E, 97) (F, 112) (G, 125) (H, 97) — D
5. (A-D-I, 110) ~~(E, 97)~~ (F, 112) (G, 125) (H, 97) — E
6. (A-B-E-K, 130) ~~(A-B-E-J, 116)~~ (A-C-F, 112) (A-C-G, 125) ~~(A-C-H, 97)~~ ~~(A-D-E, 110)~~ — H
7. (A-D-I-H, 156) (A-D-I-O, 146) ~~(A-C-F, 112)~~ ~~(A-B-E-J, 116)~~ (A-C-G, 125) (A-B-E-K, 130) — F, J
8. ~~(A-C-G-L, 90)~~ (A-C-G-M, 125), (A-B-E-K, 130) (A-D-I-O, 146) (A-D-I-N, 156) — G

L M

A B C D E H I

A-C-G-M-P, 144
A-C-GM-Q, 33

Q

ABCDE-HIF
JGLM

A - C - G - M - Q.

$H(A) \leq 33$

$40 \leq 33$ X

not admissible